A K-INJECTIVE COMPLEX OF MODULES OVER A FLAT ALGEBRA IS K-INJECTIVE OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- K-injective complex, defined by the vanishing of the group of homotopy classes of maps from any acyclic complex. defined earlier in the same section, source lines 13860-13924. Supplies the property that is transferred from the target ring to the base.
- The tensor-restriction adjunction identifying maps from a base changed complex with maps from the original complex. the author cites Algebra, Lemma algebra-lemma-adjoint-tensor-restrict at proof lines 13935-13937. Rewrites the Hom group over the target ring as a Hom group over the base.
- Flatness of the ring map makes tensoring with the target ring exact. used at proof lines 13937-13938. Preserves acyclicity so that the vanishing condition applies on the target side.

Source reference. Original lemma 0915: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11433–11433. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05DQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6963–6963. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $R \to S$ be a flat ring map. If $I^\bullet$ is a K-injective complex of S-modules, then $I^\bullet$ is K-injective as a complex of R-modules.*

Proof. This is true because $\mathrm{Hom}_{K(R)}(M^\bullet, I^\bullet) = \mathrm{Hom}_{K(S)}(M^\bullet \otimes_R S, I^\bullet)$ by Algebra, Lemma 05DQ and the fact that tensoring with S is exact. □